

Technical Design Document: Jackpot Transition Auditor (JTA)

Document Version: 1.0.0

Status: Ready for Implementation

Category: Temporal State Persistence / Compliance

1. Executive Summary

In regulated gaming jurisdictions, the transition between one jackpot cycle and the next must be near-instantaneous. A gap exceeding **5 seconds** indicates a "dark period" where player contributions may be lost or game rounds may overlap illegally. This module serves as an automated "watchdog" to assert that all transitions meet these strict temporal and state requirements.

2. Data Schema Requirements

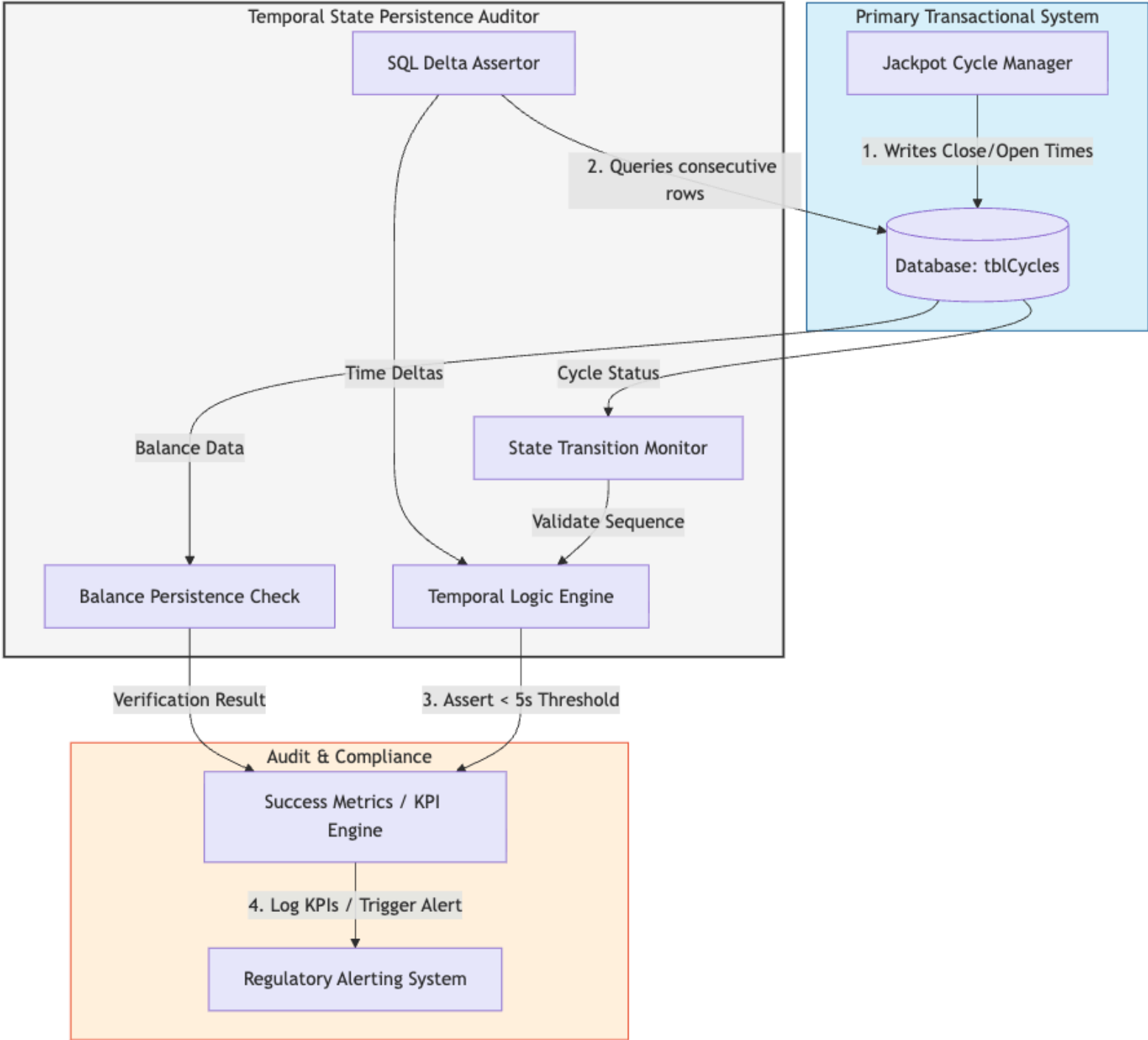
The auditor interacts primarily with the `tblCycles` table. The following fields are mandatory for the validation logic:

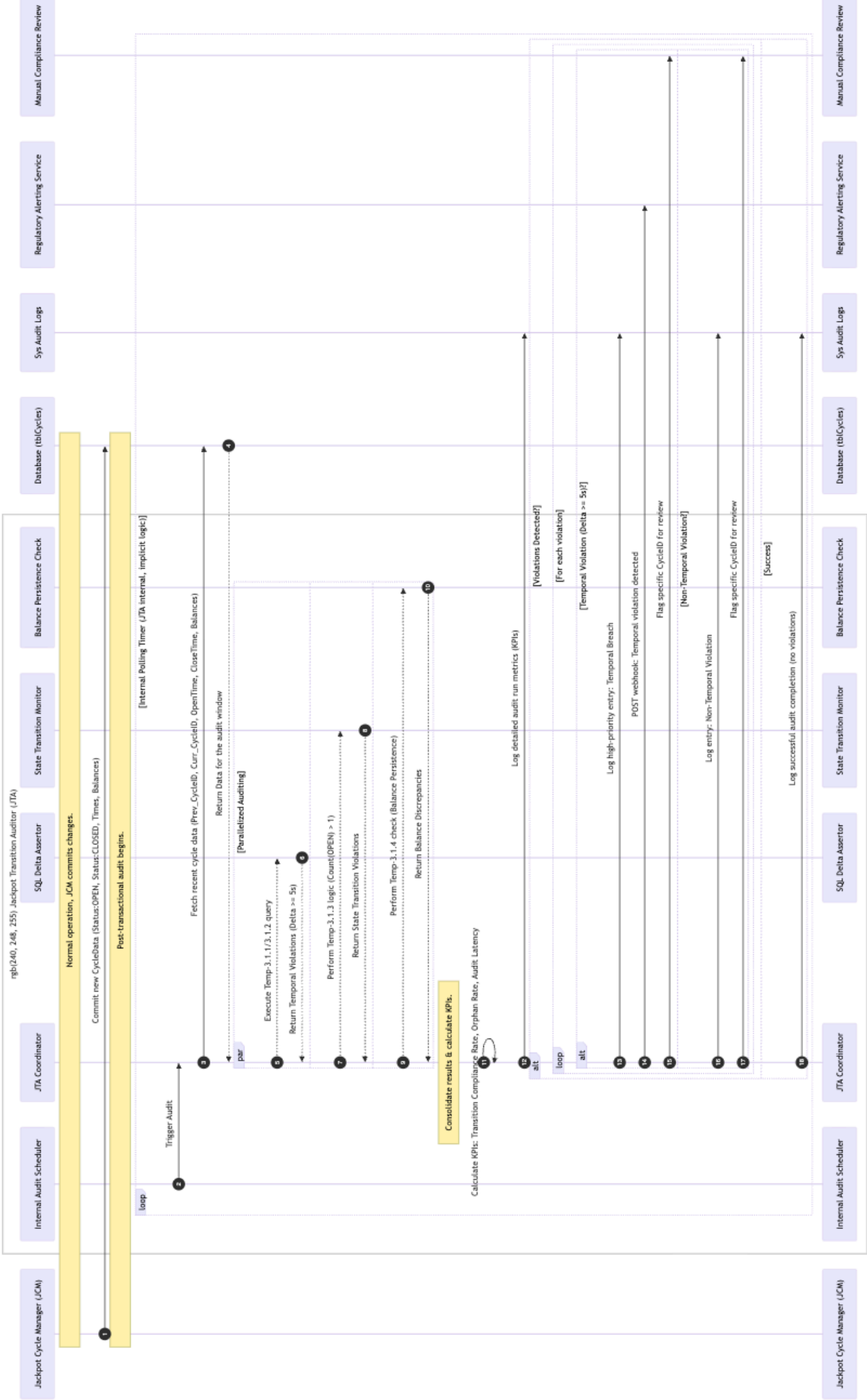
Column	Type	Description
CycleID	UUID / INT	Unique identifier for the cycle.
JackpotID	INT	Identifier for the specific jackpot pool.
Status	ENUM	Values: <code>OPEN</code> , <code>CLOSED</code> .
OpenTime	TIMESTAMP(3)	Precise time the cycle became active.
CloseTime	TIMESTAMP(3)	Precise time the cycle was terminated.
EndBalance	DECIMAL	The final monetary value of the closed cycle.

StartBalance	DECIMAL	The starting monetary value of the new cycle.
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3. System Architecture & Sequence

The JTA operates as a **post-transactional auditor**. It polls the database or listens to a Change Data Capture (CDC) stream to validate transitions after the **Jackpot Cycle Manager (JCM)** commits changes.





4. Technical Validation Components

TEMP-3.1.1 & 3.1.2: SQL Delta Assertor

The core logic utilizes SQL window functions to compare consecutive rows within the same `JackpotID`.

Logic Expression:

$\Delta = \text{OpenTime}_{\{n\}} - \text{CloseTime}_{\{n-1\}}$

Implementation Query:

SQL

None

```
SELECT
    JackpotID,
    Prev_CycleID,
    Curr_CycleID,
    EXTRACT(EPOCH FROM (OpenTime - Prev_CloseTime)) AS Delta_Seconds
FROM (
    SELECT
        JackpotID,
        CycleID AS Curr_CycleID,
        LAG(CycleID) OVER (PARTITION BY JackpotID ORDER BY OpenTime) AS Prev_CycleID,
        OpenTime,
        LAG(CloseTime) OVER (PARTITION BY JackpotID ORDER BY OpenTime) AS Prev_CloseTime
    FROM tblCycles
) AS TransitionData
WHERE Prev_CloseTime IS NOT NULL;
```

TEMP-3.1.3: State Transition Monitor

The auditor must verify that a new cycle is never instantiated while the previous cycle remains `OPEN`.

- **Assertion:** For any `JackpotID`, there can only be exactly one `OPEN` row at any given timestamp.
- **Validation:** Query for `COUNT(Status = 'OPEN') > 1`.

TEMP-3.1.4: Balance Persistence Check

To prevent "skimming" or calculation errors during transition:

- **Assertion:** $\text{EndBalance}_{\{n-1\}} + \text{SeedAmount} = \text{StartBalance}_{\{n\}}$
- **Failure Trigger:** Any discrepancy $> 0.00\$$ in base currency.

5. Success Metrics (KPIs)

Performance will be tracked via the following metrics:

1. **Transition Compliance Rate:** Percentage of cycles where $\Delta < 5s$. (Target: **100%**)

2. **Orphan Rate:** Number of `OPEN` cycles without a subsequent transition recorded. (Target: **0**)
3. **Audit Latency:** Time taken from DB commit to Audit completion. (Target: **< 10s**)

6. Error Handling & Alerting

If the **Temporal Logic Engine** detects a violation ($\Delta \geq 5s$):

1. **Log:** High-priority entry in `sys_audit_logs`.
2. **Notify:** Trigger webhook to Regulatory Alerting Service (Slack/PagerDuty).
3. **Report:** Flag the specific `CycleID` for manual compliance review.